Getabalew Manegerew Sewbekele

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EDUCATION

Addis Ababa University — Addis Ababa, Ethiopia
Bachelor of Science in Physics — **#1 Ranked in Class of 2026**

- **Graduation Date:** June 27, 2026
- **Cumulative GPA:** 4.00/4.00
- **Core Theory:** Classical Mechanics, Quantum Mechanics I & II, Electrodynamics I & II, Statistical Mechanics, Mathematical Methods I & II
- **Relevant Courses:** Astronomy and Astrophysics I & II, Computational Physics, Applications of Computational Physics, Nuclear Physics, Applications of Nuclear Physics

RESEARCH EXPERIENCE

Addis Ababa University — *Addis Ababa, Ethiopia*
Senior Thesis Project — *February – June 2026*

- **Project Title:** Direct Collisional Method of Molecular Dynamics Simulation
- **Project Theme:** Blended Direct Simulation Monte Carlo (DSMC) and Molecular Dynamics (MD) frameworks to maximize computational efficiency while fully preserving particle histories.
- Investigated elastic collision dynamics across both laboratory and center-of-mass reference frames.
- Proposed and implemented a novel computational algorithm in Fortran 90 to simulate multi-particle systems.
- Conducted qualitative and quantitative data analysis of statistical outputs using NumPy and Matplotlib.
- Demonstrated that the simulated system ergodically approaches the Maxwell-Boltzmann Distribution (MBD).
- Modeled ionizing radiation propagation in gases, validating simulation data against established theoretical predictions.

Addis Ababa University — *Addis Ababa, Ethiopia*
Undergraduate Researcher & Team Lead — *March – August 2025*

- **Project:** Investigation of atmospheric muon angular flux variation.
- Diagnosed and resolved a long-standing GPS firmware malfunction in the detector system, restoring full data acquisition capability.
- Developed a Python-based Monte Carlo simulation of muon trajectories in Earth's magnetic field, successfully reproducing experimental angular distribution data.
- Optimized detector operating voltage using the coincidence counting plateau method to drastically improve signal-to-noise performance.
- Authored a faculty-approved research proposal; currently preparing a peer-reviewed manuscript demonstrating relativistic time dilation effects utilizing low-cost muon detection systems.

TEACHING & PROFESSIONAL EXPERIENCE

Addis Ababa University — *Addis Ababa, Ethiopia*
Teaching Assistant — Statistical Mechanics — *Oct 2025 – Jan 2026*

- Delivered weekly recitation sessions to reinforce abstract statistical mechanics concepts through guided, rigorous problem-solving.

- Developed interactive computational visualizations using random number methods (Monte Carlo) to demonstrate the convergence of the Boltzmann distribution.

Ethiopian Conformity Assessment Enterprise *Addis Ababa, Ethiopia*
Radiation Spectrometry Intern *June 2025 – Sept 2025*

- Calibrated High-Purity Germanium (HPGe) gamma spectrometry systems used for national safety regulations and radiation compliance checking.
- Designed and implemented a relational database architecture using MS Access to significantly improve operational efficiency in the national dosimetry program.

SKILLS & TOOLS

- **Programming:** Python (NumPy, SciPy, Matplotlib), C++ (numerical algorithms), Fortran 90 (numerical integration, differentiation, curve fitting, RK4 solvers, Physical Modeling)
- **Software & OS:** Mathematica (symbolic computation), LaTeX (typesetting), GNUPLOT, MS Access, Linux/Unix environments
- **Hardware & Laboratory Systems:** Cosmic ray detectors, NIM electronics, HPGe spectrometers

ADVANCED SELF-DIRECTED PREPARATION

- **Advanced Core Physics (MIT OpenCourseWare):** Completed rigorous independent syllabi for Graduate-level Linear Algebra, Multivariable Calculus, Classical Mechanics, Electromagnetism, and Quantum Mechanics I.
- **Computational Physics Modeling:** Mastered advanced numerical methods via the University of California's *Python4Physics* curriculum and applied specialized techniques from *A Student's Guide to Python for Physical Modeling.*
- **General Relativity Study Group:** Active participant in an advanced, cross-university research and study cohort focusing on Schutz's *A First Course in General Relativity.* Mastered differential geometry, tensor calculus, manifolds, and currently executing spacetime geometry computations.

HONORS & SCHOLARSHIPS

- **EPSNA Merit-Based Scholarship**, Ethiopian Physics Society in North America (2025)
- **EPSNA Merit-Based Scholarship**, Ethiopian Physics Society in North America (2024)
- **Ethio-China Friendship Scholarship Award** (Highly competitive, merit-based) (2024–2025)
- **Mitsubishi Scholarship Award** (Academic excellence, merit-based) (2023–2024)
- **Selected Participant:** EPSNA Virtual Graduate-Level Summer School (Aug 2025)

SELECTED PRESENTATIONS & SPEAKING

- **Keynote Speaker**, "Physics Opportunities & Pathways," Sophomore Welcoming Event, Addis Ababa University (Oct 2025)
- **Invited Guest Lecturer**, "Revisiting the Laws of Thermodynamics," General Physics Course, Addis Ababa University (Dec 2025)
- **Seminar Speaker**, "Special Relativity and Electromagnetism," FOCAL Physics Club, Addis Ababa University (Apr 2023)

LEADERSHIP & SERVICE

FOCAL Club (Physics Student Association) *Co-Founder & Event Manager*

- Co-founded the department's premier physics-focused student organization to bridge undergraduate studies with research pathways.
- Organized bimonthly academic seminars, faculty panels, and peer student engagement events.

Class Representative *2022 – 2026*

- Served as the official liaison between students and departmental faculty; engineered a centralized digital pipeline for course material distribution and rapid academic communication.